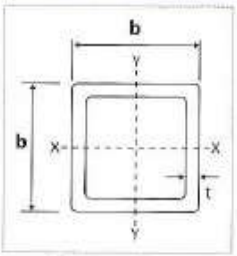
6b (i) Cold Formed Square Hollow Sections

(Australian / New Zealand Standard)
(Dimensions and Properties in accordance to AS / NZS 1163 : 2009)



Square Hollow Sections

AS/NZS 1163 : 2009

Designation	Mass Per Unit Length	External Surface Area		Ratio	Gross Area Of Cross-Section	About x, y and n - axis						
Depth x Width x Thickness		Per Unit Length	Per Unit Mass	$\frac{b-2t}{t}$		Second Moment Of Area	Elastic Section Modulus	Elastic Section Modulus	Plastic Section Modulus	Radius Of Gyration	Torsion Constant	Torsion Modulus
$b \times b \times t$	m				A_g	I_x, I_y	Z_x, Z_y	Z_n	S_x, S_y	r_x, r_y	J	C
mm x mm x mm	kg / m	m^2 / m	m^2 / t		mm^2	10^6 mm^4	10^3 mm^3	10^3 mm^3	10^3 mm^3	mm	10^6 mm^4	10^3 mm^3
20 x 20 x 1.6	0.873	0.0745	85.4	10.5	111	0.00608	0.608	0.474	0.751	7.39	0.0103	0.924
25 x 25 x 1.6	1.12	0.0945	84.1	13.6	143	0.0128	1.02	0.780	1.24	9.44	0.0212	1.54
x 2.0	1.36	0.0931	68.3	10.5	174	0.0148	1.19	0.926	1.47	9.24	0.0253	1.80
x 2.5	1.64	0.0914	55.7	8.00	209	0.0169	1.35	1.08	1.71	8.99	0.0297	2.07
x 3.0	1.89	0.0897	47.4	6.33	241	0.0184	1.47	1.21	1.91	8.74	0.0333	2.27
30 x 30 x 1.6	1.38	0.115	83.3	16.8	175	0.0231	1.54	1.16	1.84	11.5	0.0377	2.32
x 2.0	1.68	0.113	67.4	13.0	214	0.0272	1.81	1.39	2.21	11.3	0.0454	2.75
35 x 35 x 1.6	1.63	0.135	82.7	19.9	207	0.0379	2.16	1.62	2.57	13.5	0.0611	3.26
x 2.0	1.99	0.133	66.8	15.5	254	0.0451	2.58	1.95	3.09	13.3	0.0741	3.89
x 2.5	2.42	0.131	54.2	12.0	309	0.0529	3.02	2.33	3.69	13.1	0.0889	4.58
x 3.0	2.83	0.130	45.8	9.67	361	0.0595	3.40	2.67	4.23	12.8	0.102	5.18
40 x 40 x 1.6	1.88	0.155	82.3	23.0	239	0.0579	2.90	2.15	3.41	15.6	0.0927	4.36
x 2.0	2.31	0.153	66.4	18.0	294	0.0694	3.47	2.61	4.13	15.4	0.113	5.23
x 2.5	2.82	0.151	53.7	14.0	359	0.0822	4.11	3.13	4.97	15.1	0.136	6.21
x 4.0	4.09	0.143	34.9	8.00	521	0.105	5.26	4.36	6.74	14.2	0.192	8.33
50 x 50 x 1.6	2.38	0.195	81.7	29.3	303	0.117	4.68	3.44	5.48	19.6	0.185	7.03
x 2.0	2.93	0.193	65.8	23.0	374	0.141	5.66	4.20	6.66	19.5	0.226	8.51
x 2.5	3.60	0.191	53.1	18.0	459	0.169	6.78	5.09	8.07	19.2	0.275	10.2
x 3.0	4.25	0.190	44.7	14.7	541	0.195	7.79	5.92	9.39	19.0	0.321	11.8
x 4.0	5.35	0.183	34.2	10.5	681	0.229	9.15	7.33	11.4	18.3	0.403	14.3
65 x 65 x 2.0	3.88	0.253	65.3	30.5	494	0.323	9.94	7.29	11.6	25.6	0.509	14.9
x 2.5	4.78	0.251	52.6	24.0	609	0.391	12.0	8.91	14.1	25.3	0.624	18.1
x 3.0	5.68	0.250	44.1	19.7	721	0.454	14.0	10.4	16.6	25.1	0.733	21.0
75 x 75 x 2.5	5.56	0.291	52.4	28.0	709	0.614	16.4	12.0	19.1	29.4	0.971	24.6
x 3.0	6.60	0.290	43.9	23.0	841	0.716	19.1	14.2	22.5	29.2	1.15	28.7
x 3.5	7.53	0.285	37.9	19.4	959	0.797	21.3	16.1	25.3	28.8	1.32	32.5
x 4.0	8.49	0.283	33.3	16.8	1080	0.882	23.5	18.0	28.2	28.6	1.48	36.1
x 5.0	10.3	0.279	27.0	13.0	1310	1.03	27.5	21.6	33.6	28.0	1.77	42.6
x 6.0	12.0	0.274	22.8	10.5	1530	1.16	30.9	24.7	38.4	27.5	2.04	48.2
89 x 89 x 3.5	9.06	0.341	37.6	23.4	1150	1.37	30.9	23.2	36.5	34.5	2.24	47.1
x 5.0	12.5	0.334	26.7	15.8	1590	1.81	40.7	31.4	49.1	33.7	3.05	62.7
x 6.0	14.6	0.330	22.5	12.8	1870	2.06	46.2	36.3	56.6	33.2	3.54	71.6
100 x 100 x 3.0	8.96	0.390	43.5	31.3	1140	1.77	35.4	26.0	41.2	39.4	2.79	53.2
x 4.0	11.6	0.383	32.9	23.0	1480	2.23	44.6	33.5	52.6	38.8	3.63	68.0
x 5.0	14.2	0.379	26.6	18.0	1810	2.66	53.1	40.5	63.5	38.3	4.42	81.4
x 6.0	16.7	0.374	22.4	14.7	2130	3.04	60.7	47.1	73.5	37.7	5.15	93.6
x 9.0	23.5	0.361	15.4	9.11	3000	3.91	78.1	63.6	98.6	36.1	7.00	123
125 x 125 x 4.0	14.8	0.483	32.7	29.3	1880	4.52	72.3	53.6	84.5	49.0	7.25	110
x 5.0	18.2	0.479	26.3	23.0	2310	5.44	87.1	65.4	103	48.5	8.87	133
x 6.0	21.4	0.474	22.1	18.8	2730	6.29	101	76.5	120	48.0	10.4	154
x 9.0	30.6	0.461	15.1	11.9	3900	8.38	134	106	165	46.4	14.5	208
150 x 150 x 5.0	22.1	0.579	26.2	28.0	2810	9.70	129	96.1	151	58.7	15.6	197
x 6.0	26.2	0.574	22.0	23.0	3330	11.3	150	113	178	58.2	18.4	229
x 9.0	37.7	0.561	14.9	14.7	4800	15.4	205	159	248	56.6	26.1	316
200 x 200 x 5.0	29.9	0.779	26.0	38.0	3810	23.9	239	175	277	79.1	37.8	362
x 6.0	35.6	0.774	21.8	31.3	4530	28.0	280	207	327	78.6	44.8	425
x 9.0	51.8	0.761	14.7	20.2	6600	39.2	392	297	465	77.1	64.5	599
250 x 250 x 6.0	45.0	0.974	21.7	39.7	5730	56.2	450	330	521	99.0	88.7	681
x 9.0	65.9	0.961	14.6	25.8	8400	79.8	639	477	750	97.5	129	972

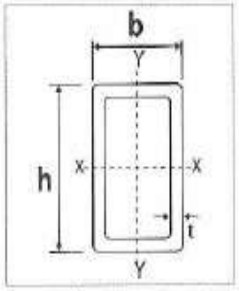
Note : The calculation of sectional properties is based on the following corner geometry:

Size Range		Inside Corner Radius mm	Outside Corner Radius mm
Thickness 3.0 mm and less	$t \le 3$ mm	1.0 t	2.0 t
Thickness greater than 3 mm	$t > 3$ mm	1.5 t	2.5 t

6b (ii) Cold Formed Rectangular Hollow Sections

(Australian / New Zealand Standard)

(Dimensions and Properties in accordance to AS / NZS 1163 : 2009)



AS/NZS 1163 : 2009

Rectangular Hollow Sections

Designation	Mass Per Unit Length	External Surface Area		Ratio		Gross Area Of Cross-section	About x - axis				About y - axis				Torsion Constant	Torsion Modulus
Depth x Width x Thickness		Per Unit Length	Per Unit Mass				Second Moment Of Area	Elastic Section Modulus	Plastic Section Modulus	Radius Of Gyration	Second Moment Of Area	Elastic Section Modulus	Plastic Section Modulus	Radius Of Gyration		
h x b x t	m			$b-2t$ / t	$h-2t$ / t	A_g	I_x	Z_x	S_x	r_x	I_y	Z_y	S_y	r_y	J	C
mm x mm x mm	kg / m	m² / m	m² / t			mm²	10^3 mm⁴	10^3 mm³	10^3 mm³	mm	10^3 mm⁴	10^3 mm³	10^3 mm³	mm	10^6 mm⁴	10^3 mm³
50 x 20 x 1.6	1.63	0.135	82.7	10.5	29.3	207	0.0608	2.43	3.14	17.1	0.0142	1.42	1.63	8.29	0.0389	2.55
x 2.0	1.99	0.133	66.8	8.00	23.0	254	0.0723	2.89	3.78	16.9	0.0167	1.67	1.96	8.11	0.0466	3.00
x 2.5	2.42	0.131	54.2	6.00	18.0	309	0.0848	3.39	4.51	16.6	0.0192	1.92	2.32	7.89	0.0550	3.49
x 3.0	2.83	0.130	45.8	4.67	14.7	361	0.0951	3.81	5.16	16.2	0.0212	2.12	2.63	7.67	0.0620	3.88
50 x 25 x 1.6	1.75	0.145	82.5	13.6	29.3	223	0.0702	2.81	3.53	17.7	0.0237	1.90	2.17	10.3	0.0585	3.29
x 2.0	2.15	0.143	66.6	10.5	23.0	274	0.0838	3.35	4.26	17.5	0.0281	2.25	2.62	10.1	0.0706	3.92
x 2.5	2.62	0.141	54.0	8.00	18.0	334	0.0989	3.95	5.11	17.2	0.0328	2.62	3.12	9.91	0.0843	4.60
x 3.0	3.07	0.140	45.5	6.33	14.7	391	0.112	4.47	5.86	16.9	0.0367	2.93	3.56	9.69	0.0964	5.18
65 x 35 x 2.0	2.93	0.193	65.8	15.5	30.5	374	0.204	6.28	7.80	23.4	0.0778	4.44	5.07	14.4	0.184	7.62
x 2.5	3.60	0.191	53.1	12.0	24.0	459	0.244	7.52	9.45	23.1	0.0926	5.29	6.13	14.2	0.223	9.10
x 3.0	4.25	0.190	44.7	9.67	19.7	541	0.281	8.65	11.0	22.8	0.106	6.04	7.11	14.0	0.259	10.4
75 x 25 x 1.6	2.38	0.195	81.7	13.6	44.9	303	0.197	5.26	6.81	25.5	0.0347	2.78	3.11	10.7	0.0993	5.05
x 2.0	2.93	0.193	65.8	10.5	35.5	374	0.238	6.36	8.31	25.3	0.0414	3.31	3.77	10.5	0.120	6.04
x 2.5	3.60	0.191	53.1	8.00	28.0	459	0.285	7.60	10.1	24.9	0.0487	3.89	4.53	10.3	0.144	7.14
75 x 50 x 2.0	3.72	0.243	65.4	23.0	35.5	474	0.372	9.91	12.0	28.0	0.199	7.96	9.06	20.5	0.414	13.1
x 2.5	4.58	0.241	52.7	18.0	28.0	584	0.450	12.0	14.6	27.7	0.240	9.60	11.0	20.3	0.505	15.9
x 3.0	5.42	0.240	44.2	14.7	23.0	691	0.522	13.9	17.1	27.5	0.278	11.1	12.9	20.0	0.593	18.4
x 4.0	6.92	0.233	33.7	10.5	16.8	881	0.630	16.8	21.1	26.7	0.335	13.4	16.0	19.5	0.754	22.7
100 x 50 x 2.0	4.50	0.293	65.1	23.0	48.0	574	0.750	15.0	18.5	36.2	0.257	10.3	11.5	21.2	0.616	17.7
x 2.5	5.56	0.291	52.4	18.0	38.0	709	0.912	18.2	22.7	35.9	0.311	12.4	14.0	20.9	0.754	21.5
x 3.0	6.60	0.290	43.9	14.7	31.3	841	1.06	21.3	26.7	35.6	0.361	14.4	16.4	20.7	0.886	25.0
x 3.5	7.53	0.285	37.9	12.3	26.6	959	1.18	23.6	29.9	35.1	0.400	16.0	18.5	20.4	1.01	28.2
x 4.0	8.49	0.283	33.3	10.5	23.0	1080	1.31	26.1	33.4	34.8	0.441	17.6	20.6	20.2	1.13	31.2
x 5.0	10.3	0.279	27.0	8.00	18.0	1310	1.53	30.6	39.8	34.1	0.511	20.4	24.4	19.7	1.35	36.5
x 6.0	12.0	0.274	22.8	6.33	14.7	1530	1.71	34.2	45.3	33.4	0.567	22.7	27.7	19.2	1.53	40.9
125 x 75 x 3.0	8.96	0.390	43.5	23.0	39.7	1140	2.43	38.9	47.3	46.1	1.11	29.5	33.3	31.1	2.43	49.5
x 4.0	11.6	0.383	32.9	16.8	29.3	1480	3.05	48.9	60.3	45.4	1.39	37.0	42.4	30.6	3.16	63.0
x 5.0	14.2	0.379	26.6	13.0	23.0	1810	3.64	58.3	72.7	44.8	1.65	43.9	51.1	30.1	3.83	75.3
150 x 50 x 3.0	8.96	0.390	43.5	14.7	48.0	1140	2.99	39.8	51.4	51.2	0.526	21.1	23.5	21.5	1.50	38.3
x 4.0	11.6	0.383	32.9	10.5	35.5	1480	3.74	49.8	65.4	50.2	0.653	26.1	29.8	21.0	1.93	48.2
x 5.0	14.2	0.379	26.6	8.00	28.0	1810	4.44	59.2	78.9	49.5	0.765	30.6	35.7	20.5	2.30	56.8
150 x 100 x 4.0	14.8	0.483	32.7	23.0	35.5	1880	5.87	78.2	94.6	55.9	3.15	63.0	71.8	40.9	6.64	105
x 5.0	18.2	0.479	26.3	18.0	28.0	2310	7.07	94.3	115	55.3	3.79	75.7	87.3	40.4	8.12	127
x 6.0	21.4	0.474	22.1	14.7	23.0	2730	8.17	109	134	54.7	4.36	87.3	102	40.0	9.51	147
200 x 100 x 4.0	17.9	0.583	32.5	23.0	48.0	2280	11.9	119	147	72.1	4.07	81.5	91.0	42.3	9.89	142
x 5.0	22.1	0.579	26.2	18.0	38.0	2810	14.4	144	179	71.5	4.92	98.3	111	41.8	12.1	172
x 6.0	26.2	0.574	22.0	14.7	31.3	3330	16.7	167	210	70.8	5.69	114	130	41.3	14.2	200
x 9.0	37.7	0.561	14.9	9.11	20.2	4800	22.8	228	293	68.9	7.64	153	180	39.9	19.9	272
250 x 150 x 5.0	29.9	0.779	26.0	28.0	48.0	3810	32.7	262	317	92.6	15.0	199	224	62.6	33.0	337
x 6.0	35.6	0.774	21.8	23.0	39.7	4530	38.4	307	374	92.0	17.5	233	264	62.2	39.0	395
x 9.0	51.8	0.761	14.7	14.7	25.8	6600	53.7	430	533	90.2	24.3	324	375	60.7	56.0	554

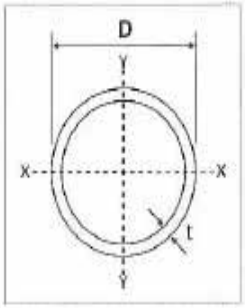
Note : The calculation of sectional properties is based on the following corner geometry:

Size Range		Inside Corner Radius mm	Outside Corner Radius mm
Thickness 3.0 mm and less	$t \le 3$ mm	1.0 t	2.0 t
Thickness greater than 3 mm	$t > 3$ mm	1.5 t	2.5 t



6b (i) Cold Formed Circular Hollow Sections

(Australian / New Zealand Standard)
(Dimensions and Properties in accordance to AS / NZS 1163 : 2009)



Circular Hollow Sections

AS/NZS 1163 : 2009

Designation				External Surface Area				About Any Axis					
Outside Diameter	x	Thickness	Unit Weight	Per Unit Length	Per Unit Mass	Ratio	Gross Area Of Cross-section	Second Moment Of Area	Elastic Section Modulus	Plastic Section Modulus	Radius Of Gyration	Torsion Constant	Torsion Modulus
D	x	t	m			D/t	A_g	I	Z	S	r	J	C
mm	x	mm	kg / m	m^2 / m	m^2 / t		mm^2	10^3 mm^4	10^3 mm^3	10^3 mm^3	mm	10^3 mm^4	10^3 mm^3
13.5	x	**2.3**	0.635	0.0424	66.8	5.9	80.9	0.00132	0.196	0.293	4.04	0.002 64	0.392
13.5	x	**2.9**	0.758	0.0424	55.9	4.7	96.6	0.00146	0.216	0.334	3.89	0.002 92	0.432
17.2	x	**2.3**	0.845	0.0540	63.9	7.5	108	0.00306	0.356	0.515	5.33	0.00612	0.711
17.2	x	**2.9**	1.02	0.0540	52.8	5.9	130	0.00347	0.403	0.601	5.16	0.00693	0.806
21.3	x	**2.6**	1.20	0.0669	55.8	8.2	153	0.00681	0.639	0.915	6.68	0.0136	1.28
21.3	x	**3.2**	1.43	0.0669	46.8	6.7	182	0.00768	0.722	1.06	6.50	0.0154	1.44
21.3	x	**3.6**	1.57	0.0669	42.6	5.9	200	0.00816	0.767	1.14	6.39	0.0163	1.53
26.9	x	**2.6**	1.56	0.0845	54.2	10.3	198	0.0148	1.10	1.54	8.64	0.0296	2.20
26.9	x	**3.2**	1.87	0.0845	45.2	8.4	238	0.0170	1.27	1.81	8.46	0.0341	2.53
26.9	x	**4.0**	2.26	0.0845	37.4	6.7	288	0.0194	1.45	2.12	8.22	0.0389	2.89
33.7	x	**3.2**	2.41	0.106	44.0	10.5	307	0.0360	2.14	2.99	10.8	0.0721	4.28
33.7	x	**4.0**	2.93	0.106	36.1	8.4	373	0.0419	2.49	3.55	10.6	0.0838	4.97
33.7	x	**4.5**	3.24	0.106	32.7	7.5	413	0.0450	2.67	3.87	10.4	0.0901	5.35
42.4	x	**3.2**	3.09	0.133	43.1	13.3	394	0.0762	3.59	4.93	13.9	0.152	7.19
42.4	x	**4.0**	3.79	0.133	35.2	10.6	483	0.0899	4.24	5.92	13.6	0.180	8.48
42.4	x	**4.9**	4.53	0.133	29.4	8.7	577	0.103	4.87	6.93	13.4	0.206	9.74
48.3	x	**3.2**	3.56	0.152	42.6	15.1	453	0.116	4.80	6.52	16.0	0.232	9.59
48.3	x	**4.0**	4.37	0.152	34.7	12.1	557	0.138	5.70	7.87	15.7	0.275	11.4
48.3	x	**5.4**	5.71	0.152	26.6	8.9	728	0.170	7.04	9.99	15.3	0.340	14.1
60.3	x	**3.6**	5.03	0.189	37.6	16.8	641	0.259	8.58	11.6	20.1	0.517	17.2
60.3	x	**4.5**	6.19	0.189	30.6	13.4	789	0.309	10.2	14.0	19.8	0.618	20.5
60.3	x	**5.4**	7.31	0.189	25.9	11.2	931	0.354	11.8	16.3	19.5	0.709	23.5
76.1	x	**2.3**	4.19	0.239	57.1	33.1	533	0.363	9.55	12.5	26.1	0.727	19.1
76.1	x	**3.2**	5.75	0.239	41.6	23.8	733	0.488	12.8	17.0	25.8	0.976	25.6
76.1	x	**3.6**	6.44	0.239	37.1	21.1	820	0.540	14.21	8.9	25.7	1.08	28.4
76.1	x	**4.5**	7.95	0.239	30.1	16.9	1010	0.651	17.1	23.1	25.4	1.30	34.2
76.1	x	**5.9**	10.2	0.239	23.4	12.9	1300	0.807	21.2	29.1	24.9	1.61	42.4
88.9	x	**2.6**	5.53	0.279	50.5	34.2	705	0.657	14.8	19.4	30.5	1.31	29.6
88.9	x	**3.2**	6.76	0.279	41.3	27.8	862	0.792	17.8	23.5	30.3	1.58	35.6
88.9	x	**4.0**	8.38	0.279	33.3	22.2	1070	0.963	21.7	28.9	30.0	1.93	43.3
88.9	x	**4.8**	9.96	0.279	28.1	18.5	1270	1.12	25.3	34.0	29.8	2.25	50.6
88.9	x	**5.0**	10.3	0.279	27.0	17.8	1320	1.16	26.2	35.2	29.7	2.33	52.4
88.9	x	**5.5**	11.3	0.279	24.7	16.2	1440	1.26	28.3	38.3	29.6	2.52	56.6
88.9	x	**5.9**	12.1	0.279	23.1	15.1	1540	1.33	30.0	40.7	29.4	2.66	59.9
101.6	x	**2.6**	6.35	0.319	50.3	39.1	809	0.991	19.5	25.5	35.0	1.98	39.0
101.6	x	**3.2**	7.77	0.319	41.1	31.8	989	1.20	23.6	31.0	34.8	2.40	47.2
101.6	x	**4.0**	9.63	0.319	33.2	25.4	1230	1.46	28.8	38.1	34.5	2.93	57.6
101.6	x	**5.0**	11.9	0.319	26.8	20.3	1520	1.77	34.9	46.7	34.2	3.55	69.9

6b) Cold-formed Structural Steel Hollow Sections–Australian/NewZealand Standard

(Extracts from AS/NZS 1163 Standard : 2009)

General Information

AS/NZS 1163 Standard specifies the requirements for cold-formed, electric resistance-welded, carbon steel hollow sections used for structural purposes. It considers three strength grades, with or without impact properties that are suitable for welding.

Chemical Composition

The Chemical analysis shall conform to the limits given in Table 1 for the appropriate grade.

Table 1. Chemical Composition

Grade	Chemical Composition (Cast Or Product Analysis) (See Note 2) % Max.										
(See Note1)	C	Si	Mn	P	S	Cr	Mo	AL (seeNote3)	Ti	Micro-Alloying Elements	CE (seeNote4)
C250, C250L0	0.12	0.05	0.50	0.03	0.03	0.15	0.10	0.10	0.04	0.03 (see Note 5)	0.25
C350, C350L0	0.20	0.45	1.60	0.03	0.03	0.30	0.10	0.10	0.04	0.15 (see Note 6)	0.43
C450, C450L0	0.20	0.45	1.70	0.03	0.03	0.50	0.35	0.10	0.04	0.15 (see Note 6)	0.43

Notes:

1. The use of sulphide modification manufacturing techniques for these grades is permitted.
2. The following elements may be present to the limits stated:
   a) Copper 0.25%
   b) Nickel 0.25%
3. Limits specified are for soluble or total aluminium.
4. Applies to niobium and vanadium only. However, niobium greater than 0.010% is not permitted.
5. Applies to niobium, vanadium and titanium only. However, vanadium greater than 0.10% is not permitted.

Mechanical Strength (TensileTest)

For Tensile test. The yield strength, tensile strength and elongation of the test piece shall conform to the limits given in Table 2 for the appropriate grade.

Table 2. Tensile Test Requirements

Grade	Minimum Yield Strength MPa	Minimum Tensile Strength MPa	Minimum Elongation As A Proportion Of The Gauge Length Of 5.65 √ So %					
			Circular Hollow Sections do/t			Rectangular Hollow Sections b/t, d/t		
			≤ 15	> 15 ≤ 30	> 30	≤ 15	> 15 ≤ 30	> 30
C250, C250L0	250	320	18	20	22	14	16	18
C350, C350L0	350	430	16	18	20	12	14	16
C450, C450L0	450	500	12	14	16	10	12	14

Impact Test

The absorbed energy value (J) resulting from–

a) each individual test: and

b) the average of three tests,

Shall conform to the specified values given in Table 3 for the appropriate grade.

Table 3: Charpy V-Notch Impact Test Requirements

Grade	Test Temperature °C	Minimum Absorbed Energy, Joule					
		Size Of Test Piece					
		10 mm X 10 mm		10 mm X 7.5 mm		10 mm X 5 mm	
		Average Of 3 Tests	Individual Test	Average Of 3 Tests	Individual Test	Average Of 3 Tests	Individual Test
C 250 L 0 C 350 L 0 C 450 L 0	0	27	20	22	16	18	13

Cold Flattening Test

The test piece shall be cut in the transverse direction and the length of the test piece shall be not less than 40mm. The flattening test piece shall be flattened at room temperature between two parallel plane surfaces with the weld seam located as follows in relation to the direction of flattening :

a) For do ≤ 60 mm: 45 degrees

b) For do > 60mm : 90 degrees

The test piece shall be flattened until the distance between the surfaces is 0.75 do or less. The test piece taken from a circular section with a longitudinal weld show no signs of cracks or flaws.

Tolerances On Dimensions and Mass

Tolerances and limits on the dimensions and mass of cold-formed hollow sections shall conform to the values given in Table 4.

Table 4. Tolerance on dimensions

Characteristic	Circular Hollow Sections	Square And Rectangular Hollow Sections	
Outside Dimension *(OD)*	± 1%, with a minimum of ± 0.5 mm and a maximum of ± 10 mm	± 1%, with minimum of ± 0.50 mm	
Thickness *(t)*	For do ≤ 406.4 mm : ± 10 % For do > 406.4 mm : ± 10 % with a max of ± 2 mm	± 10%	
Length	<table><tr><th>Type Of Length</th><th>Range mm</th><th>Tolerance</th></tr><tr><td>Random length</td><td>4000 to 16000 with a range of 2000 per order item</td><td>10 % of sections supplied may be below the minimum for the ordered range but not lesst han 75 % of the minimum</td></tr><tr><td>Mill (or' unspecified') length</td><td>All</td><td>+ 100 mm 0</td></tr><tr><td>Precision length</td><td>< 6000 ≥ 6000 ≤ 10000 > 10000</td><td>+ 5 mm 0 + 15 mm 0 + 5 mm + 1mm/m 0</td></tr></table>		
Straightness	0.20 % of total length	0.15 % of total length	
Out-of-roundness *(o)*	2 % for hollow sections having a diameter to thickness ratio not exceeding 100	-	
Concavity / convexity	-	Max. 0.8 % or 0.50 mm, whichever is greater	
Radius of Corners	-	Perimeter mm	External Corner Profile mm
		Equivalent to 50 x 50 or less	1.5 t to 3.0 t
		Equivalent to greater than 50 x 50	1.8 t to 3.0 t
Squareness of sides		90° ± 1°	
Twist *(v)*		2 mm plus 0.5 mm /m length	
Inner Flash	-		
End tolerance on diameter	-		
End Facing	-		
Mass *(m)* per unit length	Not less than 0.96 times the specified mass on individual lengths		